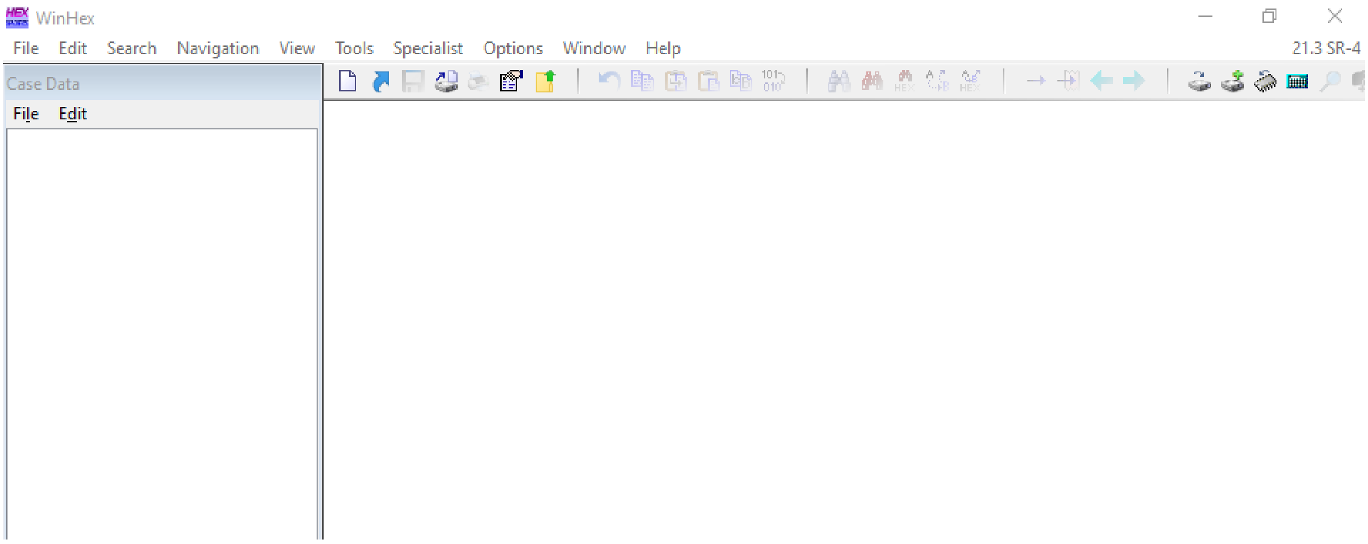
Cyber Forensics - 24CY611
Lab 2 - Understanding Hard Disks and File Systems

1. Recovering Deleted Files from Hard Disks Using WinHex

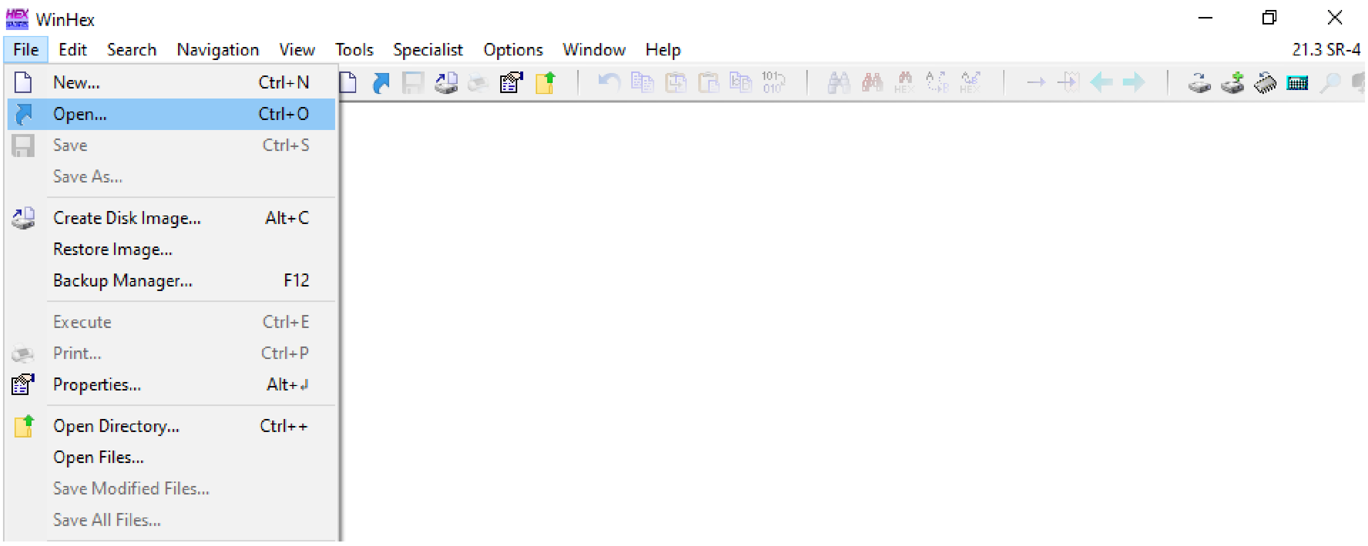
WinHex

WinHex is a versatile hexadecimal editor used for data recovery, digital forensics, and low-level file or disk analysis. It allows users to inspect and edit raw data on storage devices, memory, or files in hex format.

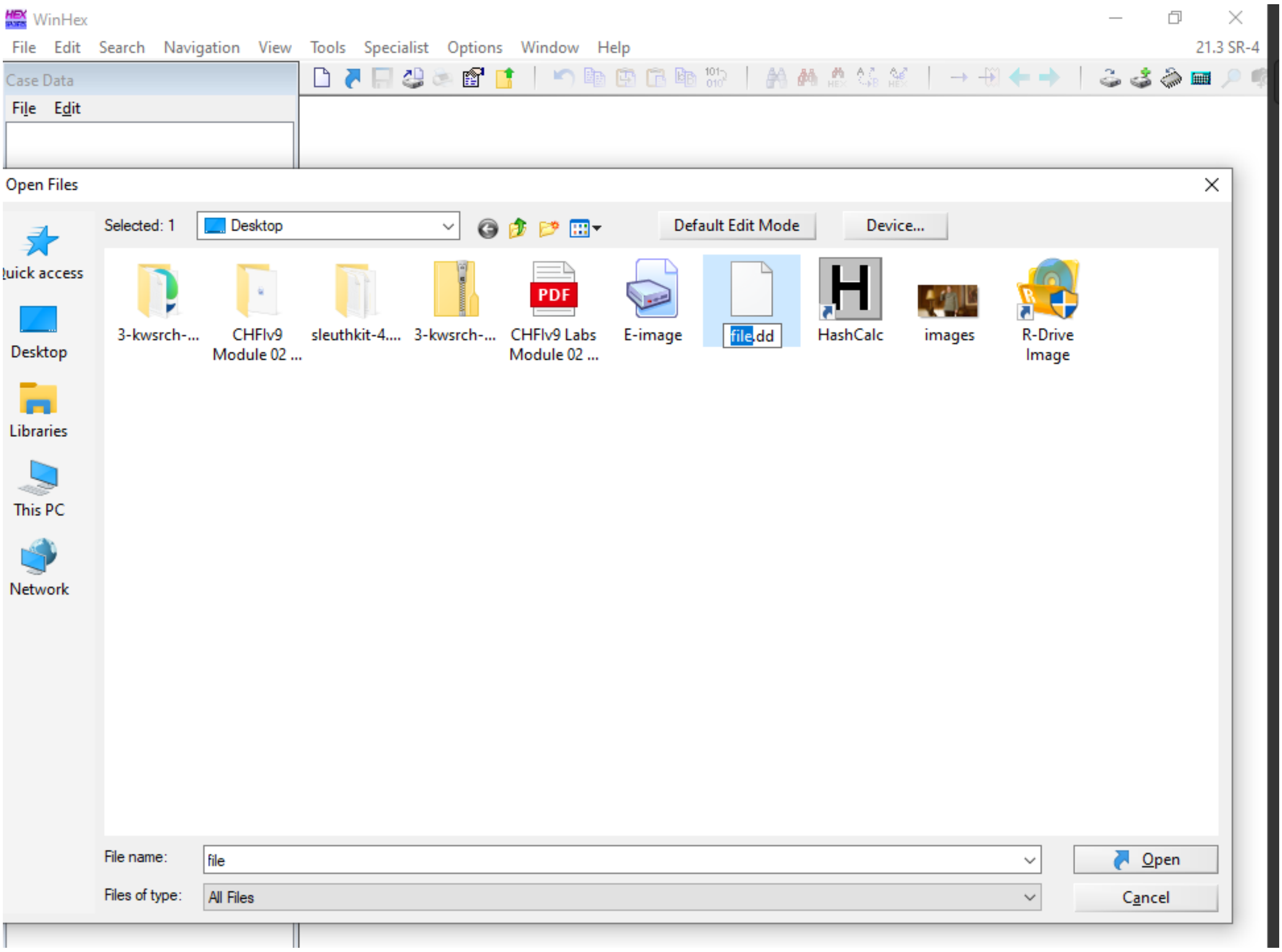
1. Install and open WinHex



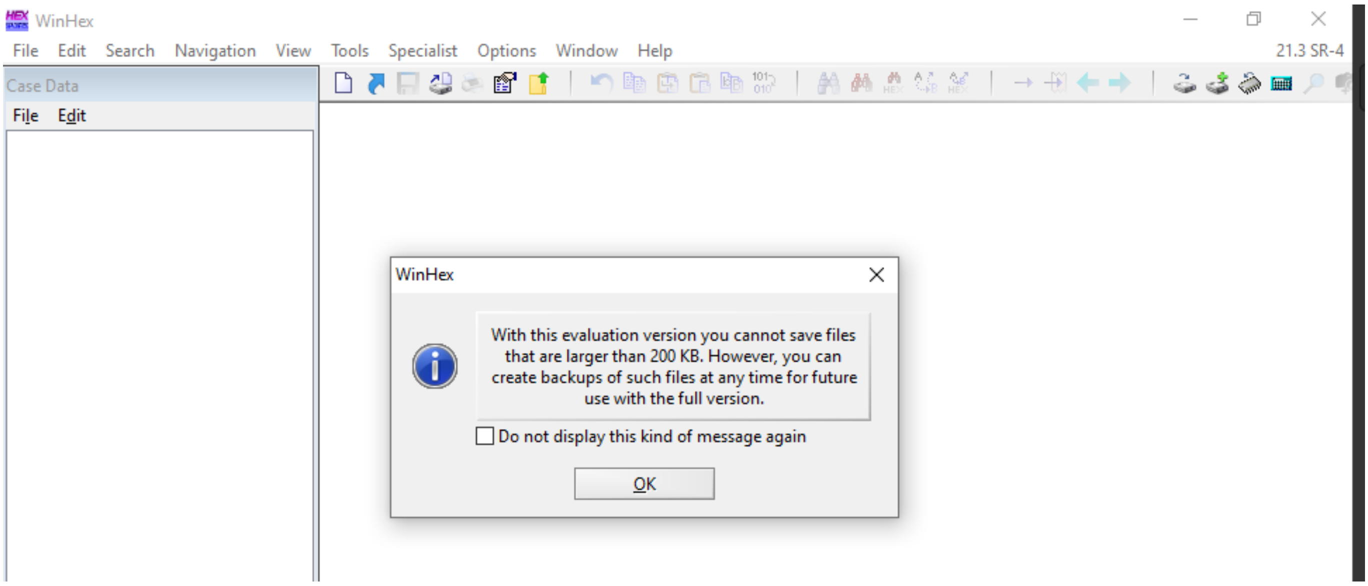
2. Navigate to File and click Open to add the evidence file



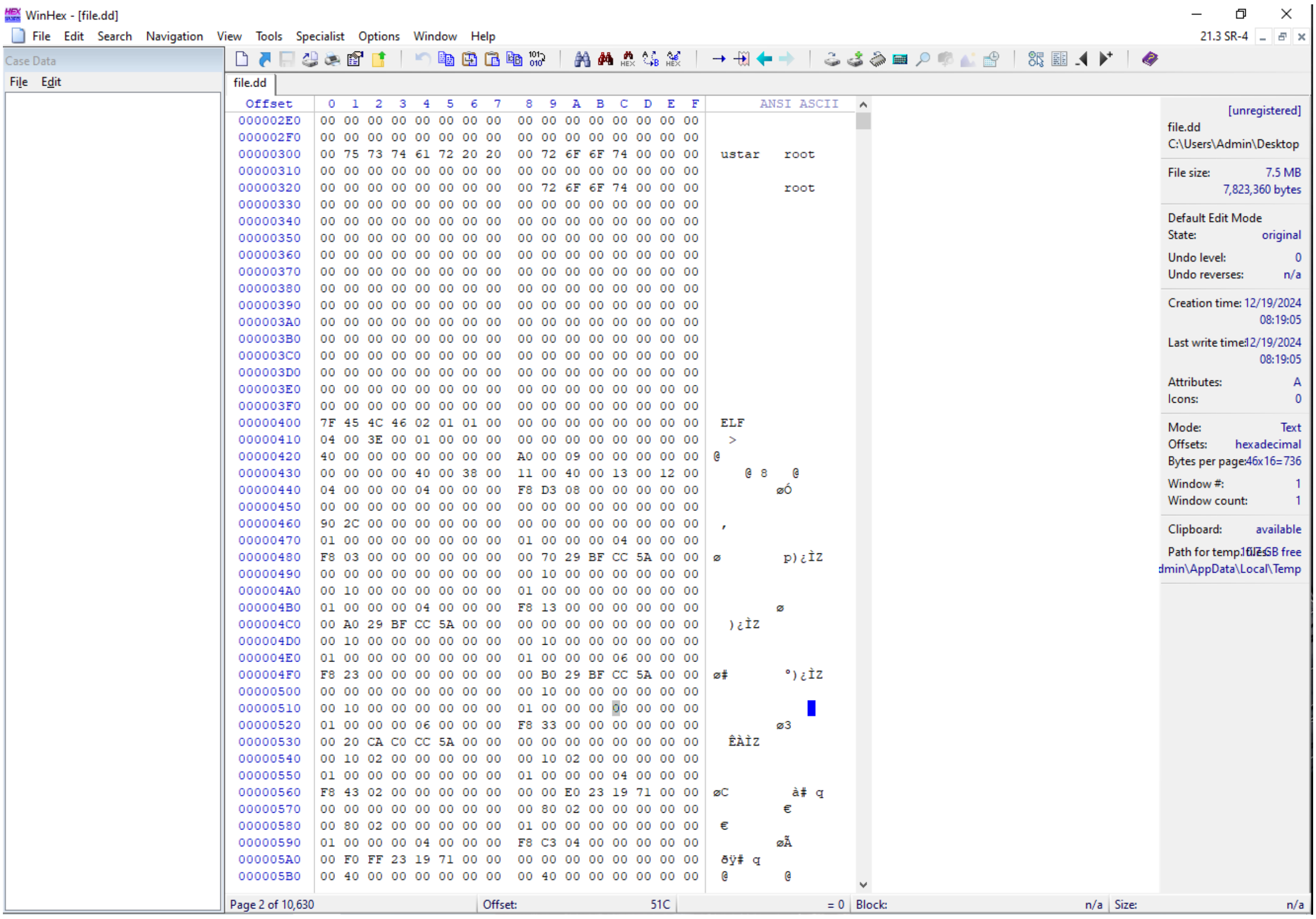
3. Then select the disk image file from the file explorer. Next click open .



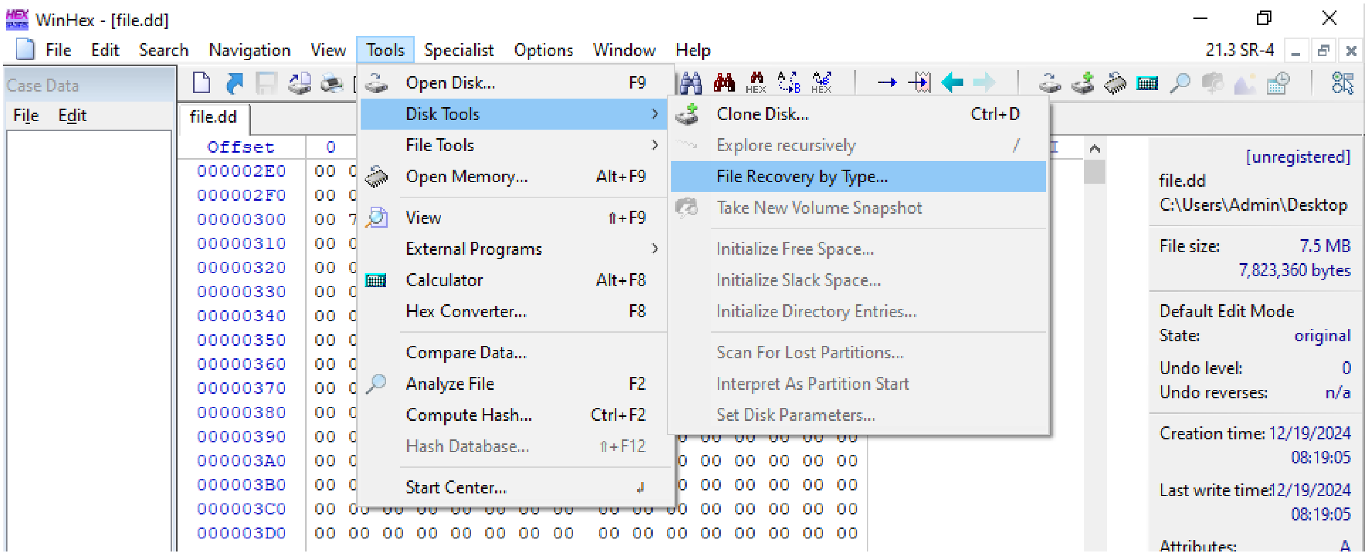
4. Then WinHex evaluation popup appears , click OK



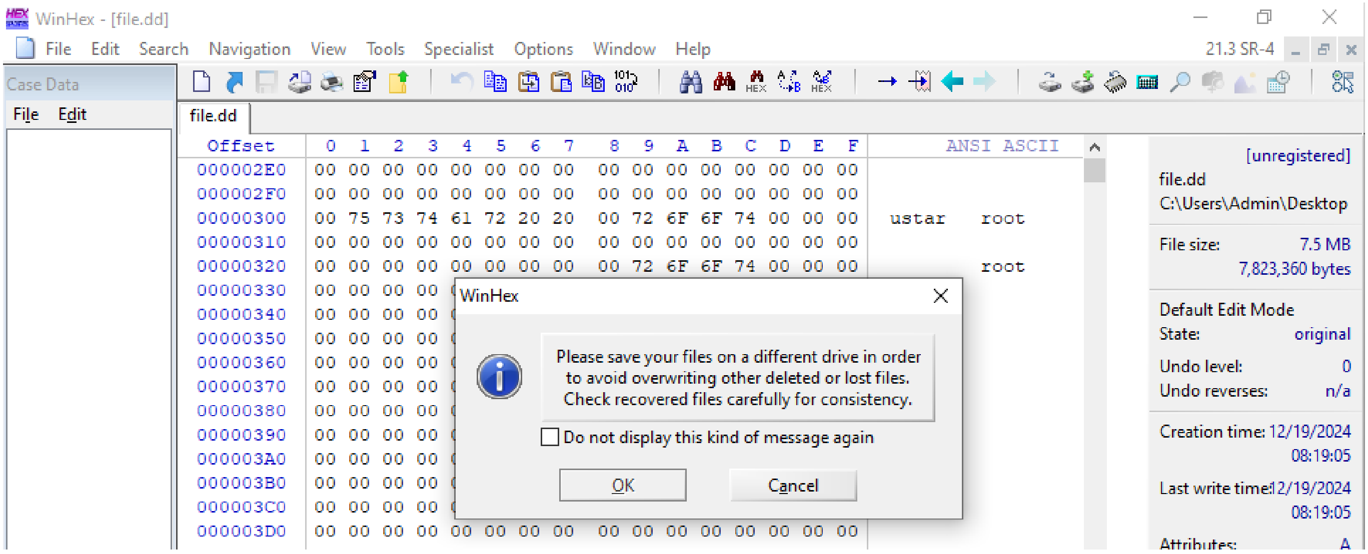
5. WinHex will process the image file and display the data as hexadecimal values in a Data Interpreter.



6. Navigate to Tools → Disk Tools → File Recovery by Type

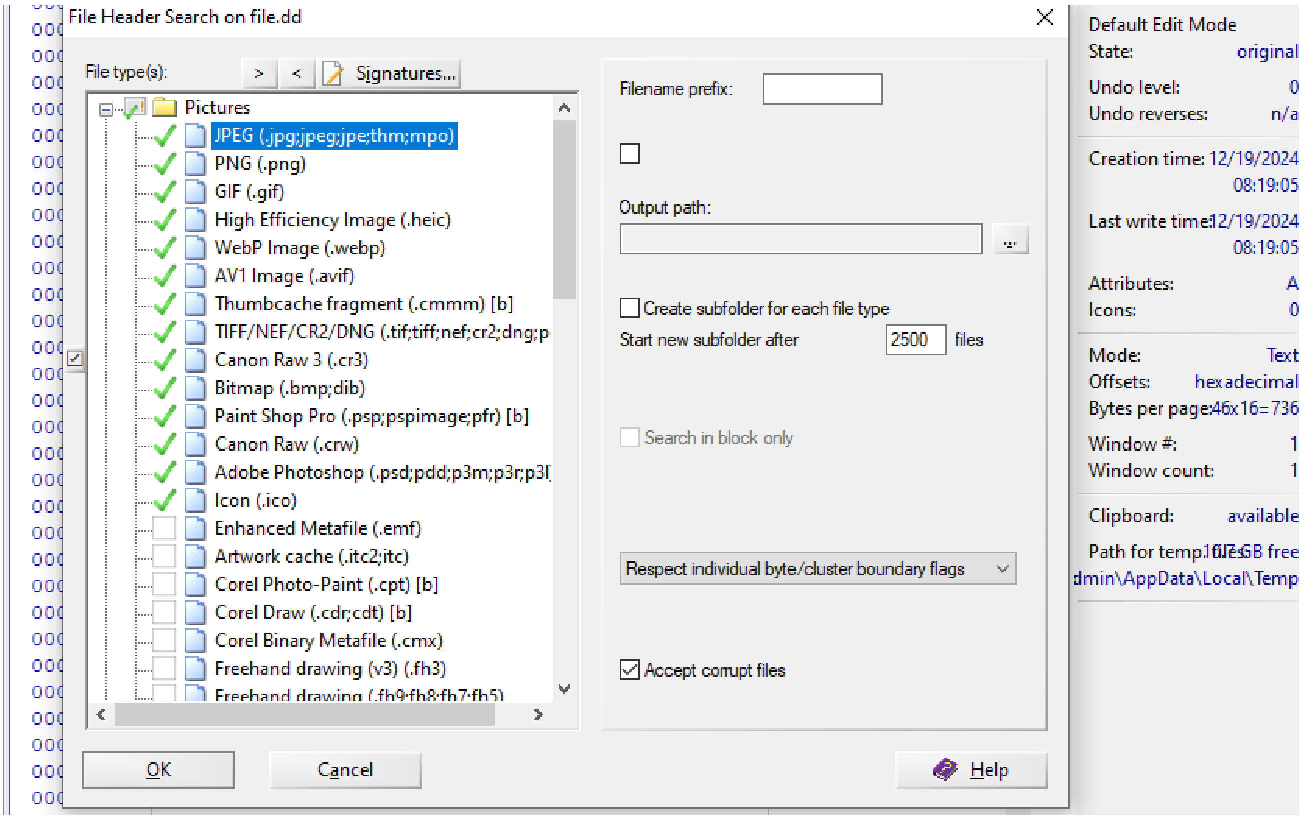


7. Click OK in the popup

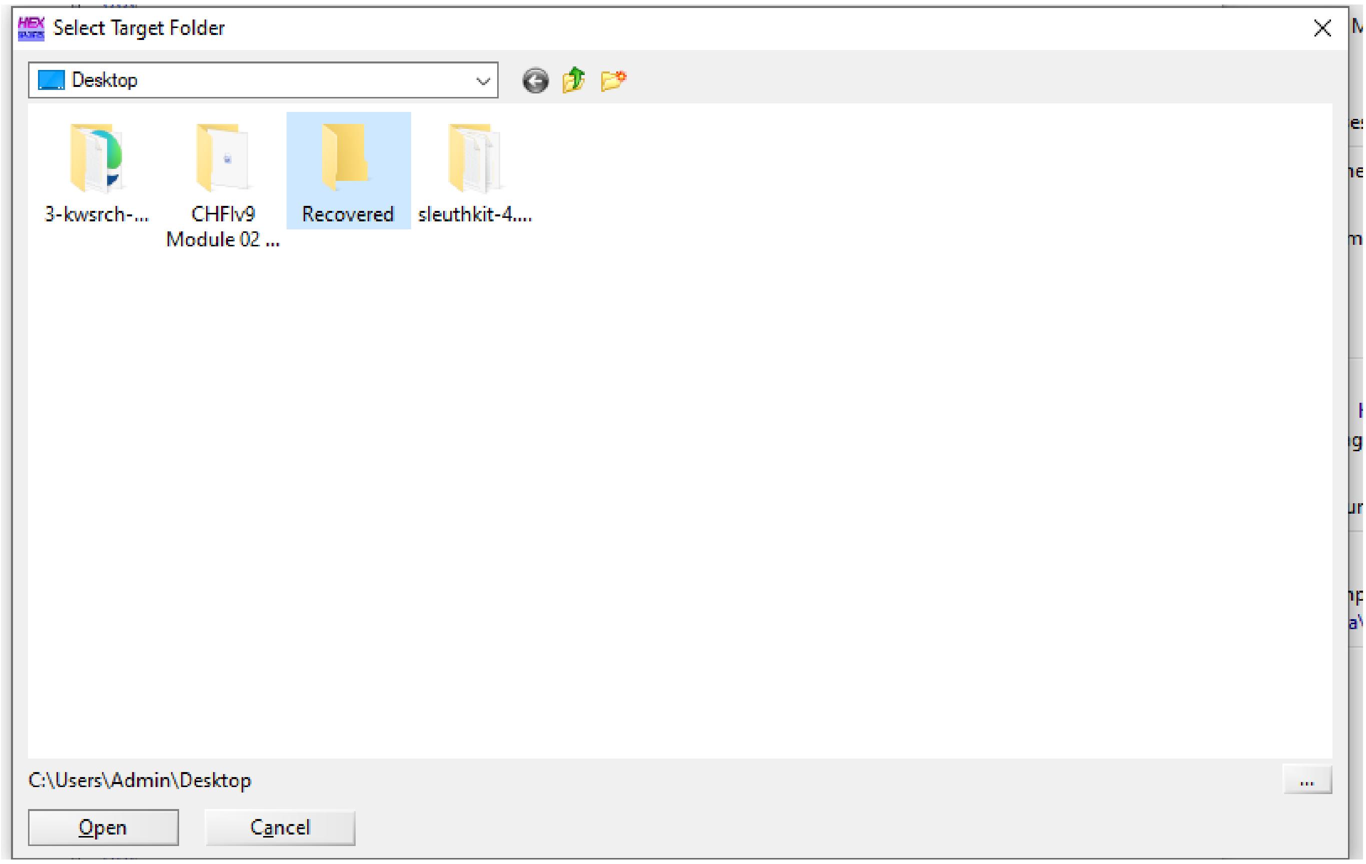


8. File header search on file.dd window appears. In this lab we are going to extract the pictures folder , click on + node to expand the pictures folder.

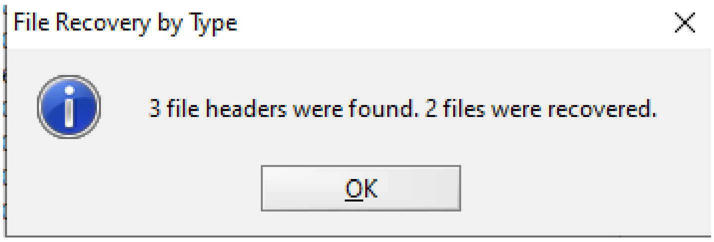
9. Select the file types of the target files to recover and then click OK.



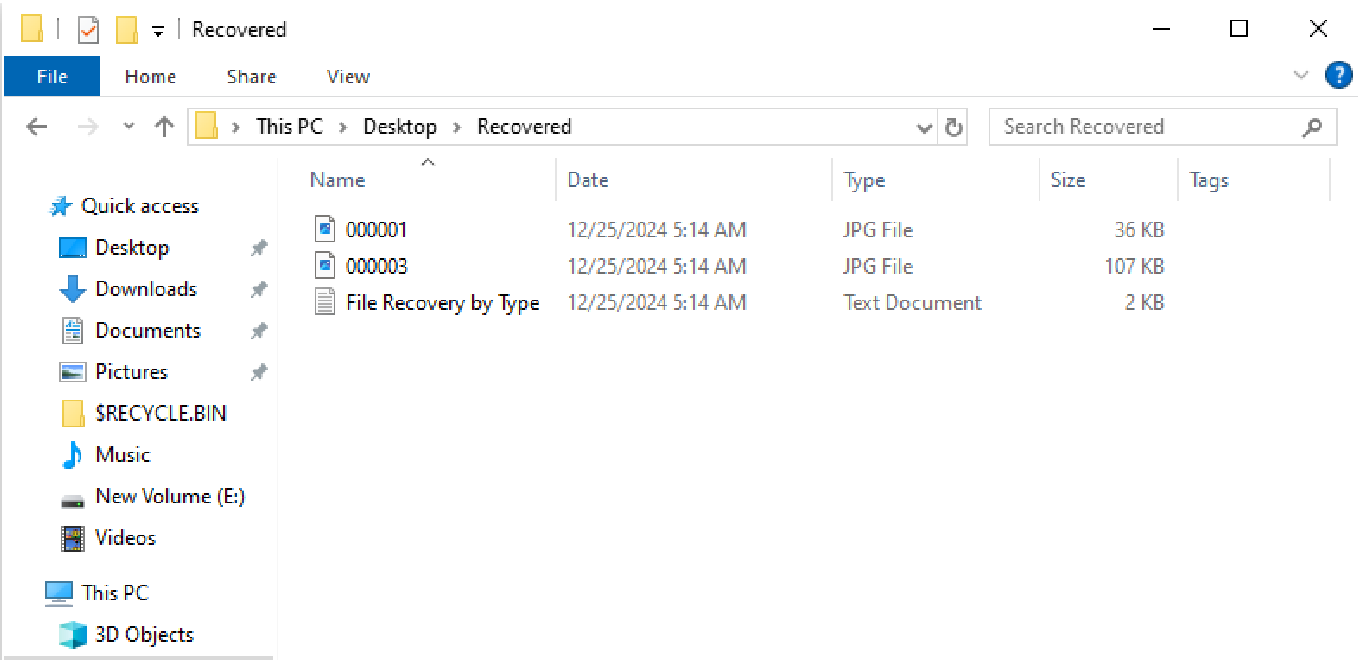
10. Select the target folder to save the recovered files and then click OK.



11. After recovery process is complete , click OK in the popup.



12. Open the target folder to see the recovered files.

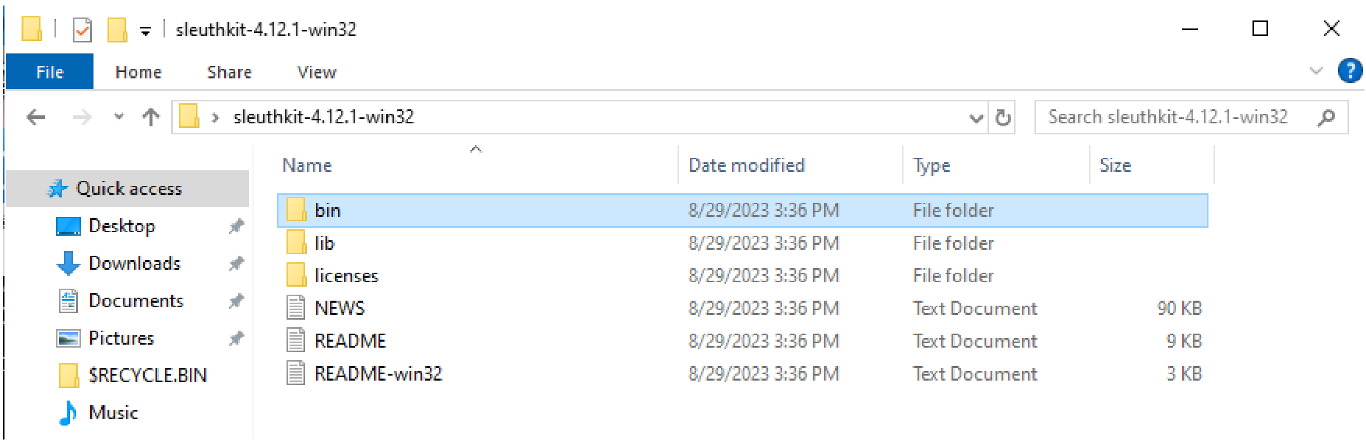


2. Analyzing File System Types Using The Sleuth Kit (TSK)

The Sleuth Kit

The Sleuth Kit (TSK) is an open-source digital forensics toolkit for analyzing file systems and recovering data. It supports various file systems like NTFS, FAT, and ext, enabling investigators to examine partitions, metadata, and deleted files. TSK's command-line tools provide detailed insights into file system structures and activities for forensic analysis.

1. Navigate to **C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin** and open with command prompt.



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>
```

2. Now type **fsstat -f “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd”** and then press Enter to see the file system details.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fsstat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: NTFS
Volume Serial Number: BA50ED9250ED5623
OEM Name: NTFS
Volume Name: KW-SRCH-1
Version: Windows XP

METADATA INFORMATION
--------------------------------------------
First Cluster of MFT: 5355
First Cluster of MFT Mirror: 8032
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 39
Root Directory: 5

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 16063
Total Sector Range: 0 - 16063

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)   Size: 48-72   Flags: Resident
$ATTRIBUTE_LIST (32)   Size: No Limit   Flags: Non-resident
$FILE_NAME (48)   Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)   Size: 0-256   Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit   Flags: Non-resident
$VOLUME_NAME (96)   Size: 2-256   Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)   Size: No Limit   Flags:
$INDEX_ROOT (144)   Size: No Limit   Flags: Resident
$INDEX_ALLOCATION (160)   Size: No Limit   Flags: Non-resident
$BITMAP (176)   Size: No Limit   Flags: Non-resident
$REPARSE_POINT (192)   Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)   Size: 8-8   Flags: Resident
$EA (224)   Size: 0-65536   Flags:
$LOGGED_UTILITY_STREAM (256)   Size: 0-65536   Flags: Non-resident
```

3. Use the istat tool to view the details of metadata structure.

4. Now type **fsstat -f “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd” 0** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 0
MFT Entry Header Values:
Entry: 0        Sequence: 1
$LogFile Sequence Number: 1075381
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256  (S-1-5-21-1757981266-484763869-1060284298-1003)
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $MFT
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 16384           Actual Size: 16384
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-3)   Name: N/A   Resident   size: 74
Type: $DATA (128-1)   Name: N/A   Non-Resident   size: 39936  init_size: 39936
5355 5356 5357 5358 5359 5360 5361 5362
5363 5364 5365 5366 5367 5368 5369 5370
5371 5372 5373 5374 5375 5376 5377 5378
5379 5380 5381 5382 5383 5384 5385 5386
5387 5388 5389 5390 5391 5392 5393 5394
5395 5396 5397 5398 5399 5400 5401 5402
5403 5404 5405 5406 5407 5408 5409 5410
5411 5412 5413 5414 5415 5416 5417 5418
5419 5420 5421 5422 5423 5424 5425 5426
5427 5428 5429 5430 5431 5432 5433 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
Type: $BITMAP (176-5)   Name: N/A   Non-Resident   size: 8  init_size: 8
5354
```

5.To view MFTMirr File Overview, type **fsstat -f “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd” 1** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 1
MFT Entry Header Values:
Entry: 1        Sequence: 1
$LogFile Sequence Number: 1052784
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256  (S-1-5-21-1757981266-484763869-1060284298-1003)
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $MFTMirr
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 4096    Actual Size: 4096
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-2)   Name: N/A   Resident   size: 82
Type: $DATA (128-1)   Name: N/A   Non-Resident   size: 4096  init_size: 4096
8032 8033 8034 8035 8036 8037 8038 8039
```

The Master File Table (MFT) is a critical component of the NTFS file system, storing metadata about every file and directory on a disk. Each record in the MFT contains details such as file names, timestamps, permissions, and physical location on the storage device

6. To view Boot File Overview, type **fsstat -f “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd” 7** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 7
MFT Entry Header Values:
Entry: 7        Sequence: 7
$LogFile Sequence Number: 0
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 0  ()
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $Boot
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 8192    Actual Size: 8192
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 48
Type: $FILE_NAME (48-2)   Name: N/A   Resident   size: 76
Type: $SECURITY_DESCRIPTOR (80-3)   Name: N/A   Resident   size: 116
Type: $DATA (128-1)   Name: N/A   Non-Resident   size: 8192  init_size: 8192
0 1 2 3 4 5 6 7
8 9 10 11 12 13 14 15
```

7. To view File Volume Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 3** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 3
MFT Entry Header Values:
Entry: 3        Sequence: 3
$LogFile Sequence Number: 1087457
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 0  ()
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $Volume
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 0       Actual Size: 0
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: 8edb73da-63d1-4dcd-b2d9-7035350ef062

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 48
Type: $FILE_NAME (48-1)   Name: N/A   Resident   size: 80
Type: $OBJECT_ID (64-6)   Name: N/A   Resident   size: 16
Type: $SECURITY_DESCRIPTOR (80-2)   Name: N/A   Resident   size: 116
Type: $VOLUME_NAME (96-4)   Name: N/A   Resident   size: 18
Type: $VOLUME_INFORMATION (112-5)   Name: N/A   Resident   size: 12
Type: $DATA (128-3)   Name: N/A   Resident   size: 0
```

8. To viewAttrDef File Overview, type **fsstat -f "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 4** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 4
MFT Entry Header Values:
Entry: 4        Sequence: 4
$LogFile Sequence Number: 1053965
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 0  ()
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $AttrDef
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 36352           Actual Size: 36000
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 48
Type: $FILE_NAME (48-2)   Name: N/A   Resident   size: 82
Type: $SECURITY_DESCRIPTOR (80-3)   Name: N/A   Resident   size: 116
Type: $DATA (128-4)   Name: N/A   Non-Resident   size: 2560  init_size: 2560
5339 5340 5341 5342 5343
```

9. To view Bitmap File Overview, type **fsstat -f “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd” 6** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 6
MFT Entry Header Values:
Entry: 6        Sequence: 6
$LogFile Sequence Number: 1052934
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256  (S-1-5-21-1757981266-484763869-1060284298-1003)
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $Bitmap
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 2048    Actual Size: 2008
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-2)   Name: N/A   Resident   size: 80
Type: $DATA (128-1)   Name: N/A   Non-Resident   size: 2008  init_size: 2008
8128 8129 8130 8131

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>
```

10. To view BiadClus File Overview, type **fsstat -f “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd” 8** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 8
MFT Entry Header Values:
Entry: 8        Sequence: 8
$LogFile Sequence Number: 1053084
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 256  (S-1-5-21-1757981266-484763869-1060284298-1003)
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Hidden, System
Name: $BadClus
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 0       Actual Size: 0
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-3)   Name: N/A   Resident   size: 82
Type: $DATA (128-2)   Name: N/A   Resident   size: 0
Type: $DATA (128-1)   Name: $Bad   Non-Resident   size: 8224768  init_size: 0

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>
```

11. To view Secure File Overview, type **fsstat -f “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd” 9** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>istat -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd" 9
MFT Entry Header Values:
Entry: 9        Sequence: 9
$LogFile Sequence Number: 1086357
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Hidden, System
Owner ID: 0
Security ID: 257  (S-1-5-21-1757981266-484763869-1060284298-1003)
Created:        2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
File Modified:  2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
MFT Modified:   2003-10-23 10:12:59.693550400 (Pacific Daylight Time)
Accessed:       2003-10-23 10:12:59.693550400 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags:
Name: $Secure
Parent MFT Entry: 5     Sequence: 5
Allocated Size: 0       Actual Size: 0
Created:        2076-11-29 00:54:34.000000000 (Pacific Standard Time)
File Modified:  2076-11-29 00:54:34.000000000 (Pacific Standard Time)
MFT Modified:   2076-11-29 00:54:34.000000000 (Pacific Standard Time)
Accessed:       2076-11-29 00:54:34.000000000 (Pacific Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-7)   Name: N/A   Resident   size: 80
Type: $DATA (128-8)   Name: $SDS   Non-Resident   size: 264040  init_size: 264040
16 17 18 19 20 21 22 23
24 25 26 27 28 29 30 31
32 33 34 35 36 37 38 39
40 41 42 43 44 45 46 47
48 49 50 51 52 53 54 55
56 57 58 59 60 61 62 63
64 65 66 67 68 69 70 71
72 73 74 75 76 77 78 79
80 81 82 83 84 85 86 87
88 89 90 91 92 93 94 95
96 97 98 99 100 101 102 103
104 105 106 107 108 109 110 111
112 113 114 115 116 117 118 119
120 121 122 123 124 125 126 127
128 129 130 131 132 133 134 135
136 137 138 139 140 141 142 143
144 145 146 147 148 149 150 151
152 153 154 155 156 157 158 159
160 161 162 163 164 165 166 167
168 169 170 171 172 173 174 175
```

12. To vlist files and directory, type **fls -f “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd”** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fls -f ntfs "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"
r/r 4-128-4:    $AttrDef
r/r 8-128-2:    $BadClus
r/r 8-128-1:    $BadClus:$Bad
r/r 6-128-1:    $Bitmap
r/r 7-128-1:    $Boot
d/d 11-144-4:   $Extend
r/r 2-128-1:    $LogFile
r/r 0-128-1:    $MFT
r/r 1-128-1:    $MFTMirr
r/r 9-128-8:    $Secure:$SDS
r/r 9-144-11:   $Secure:$SDH
r/r 9-144-14:   $Secure:$SII
r/r 10-128-1:   $UpCase
r/r 3-128-3:    $Volume
d/r 38-128-4:   dir-n-6:there
d/d 38-144-1:   dir-n-6
d/r 30-128-3:   dir-r-4:there
d/d 30-144-1:   dir-r-4
r/r 33-128-3:   file-n-1.dat
r/r 35-128-3:   file-n-3.dat
r/r 36-128-3:   file-n-4.dat
r/r 37-128-3:   file-n-5.dat
r/r 37-128-5:   file-n-5.dat:here
r/r 27-128-1:   file-r-1.dat
r/r 29-128-1:   file-r-3.dat
r/r 29-128-3:   file-r-3.dat:here
d/d 31-144-1:   System Volume Information
-/r * 34-128-1: file-r-2.dat
V/V 39: $OrphanFiles
```

13. To view only deleted entries, type **fls -d “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd”** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>fls -d "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"
-/r * 34-128-1: file-r-2.dat

C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>
```

14. To view details of an image file, type **img_stat “C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd”** and then press Enter.

```
C:\Users\Admin\Desktop\sleuthkit-4.12.1-win32\bin>img_stat "C:\Users\Admin\Desktop\3-kwsrch-ntfs\ntfs-img-kw-1.dd"
IMAGE FILE INFORMATION
--------------------------------------------
Image Type: raw

Size in bytes: 8224768
Sector size:    512
```

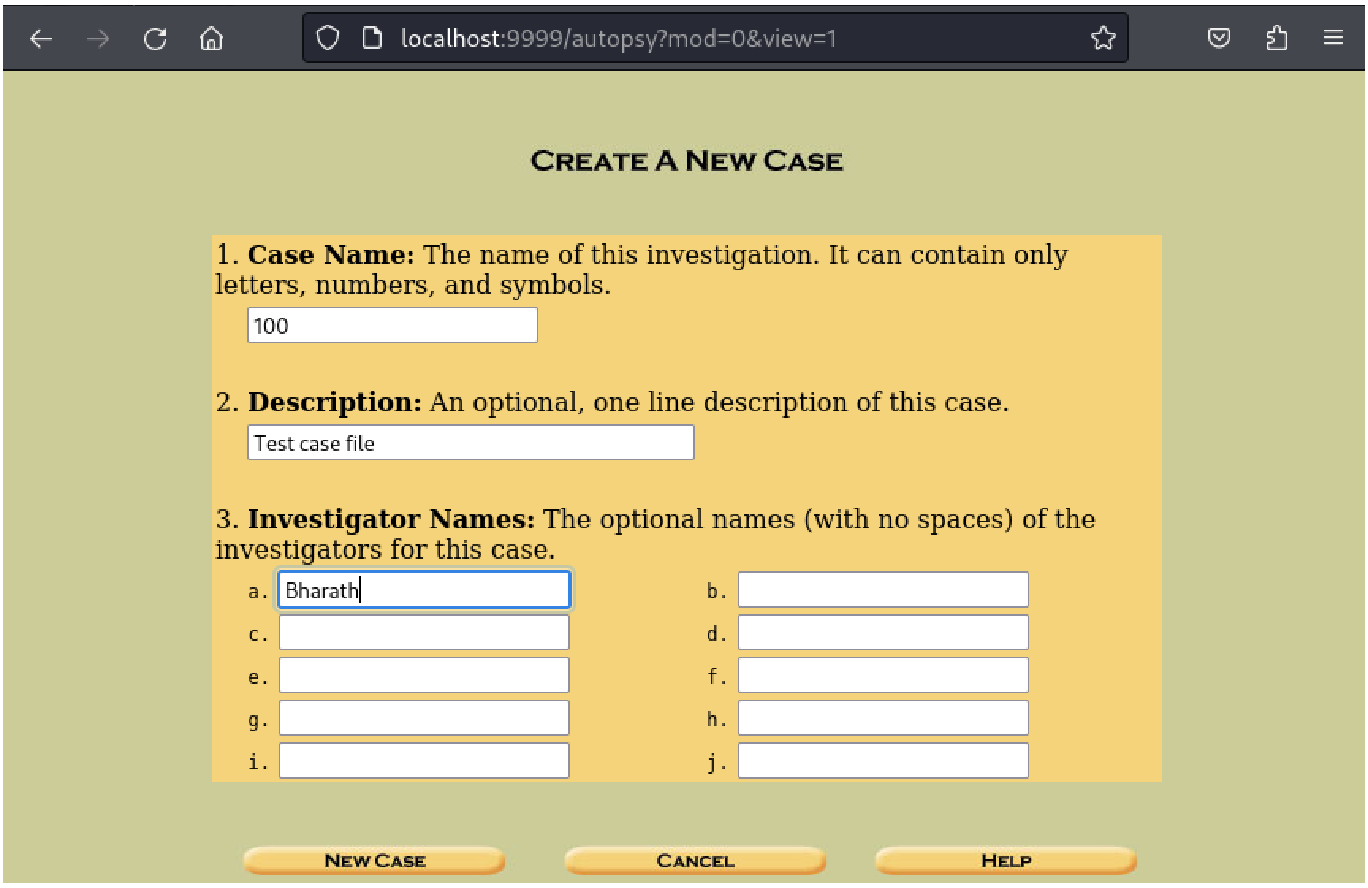
3. Analysing Raw Image using Autopsy

Autopsy is a user-friendly, open-source digital forensics platform for investigating data breaches, recovering deleted files, and analyzing disk images.

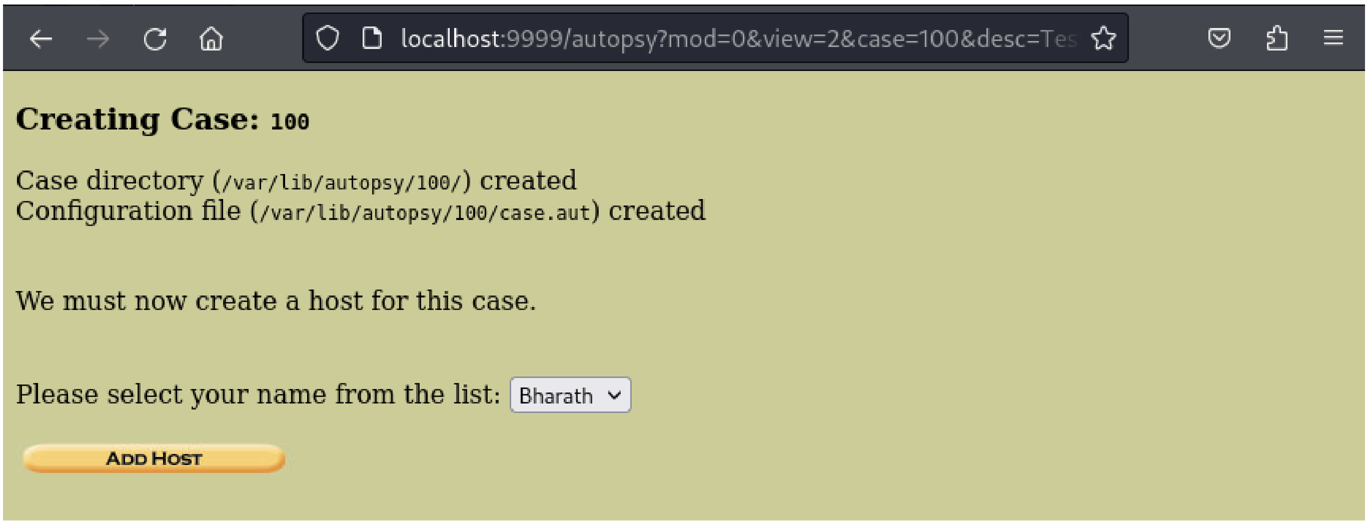
1. Open terminal in kali type sudo autopsy to start autopsy.

2. Then copy the link http://localhost:9999/autopsy in the terminal to browser. On loading the url autopsy main window will appear in the browser. Click **NEW CASE.**

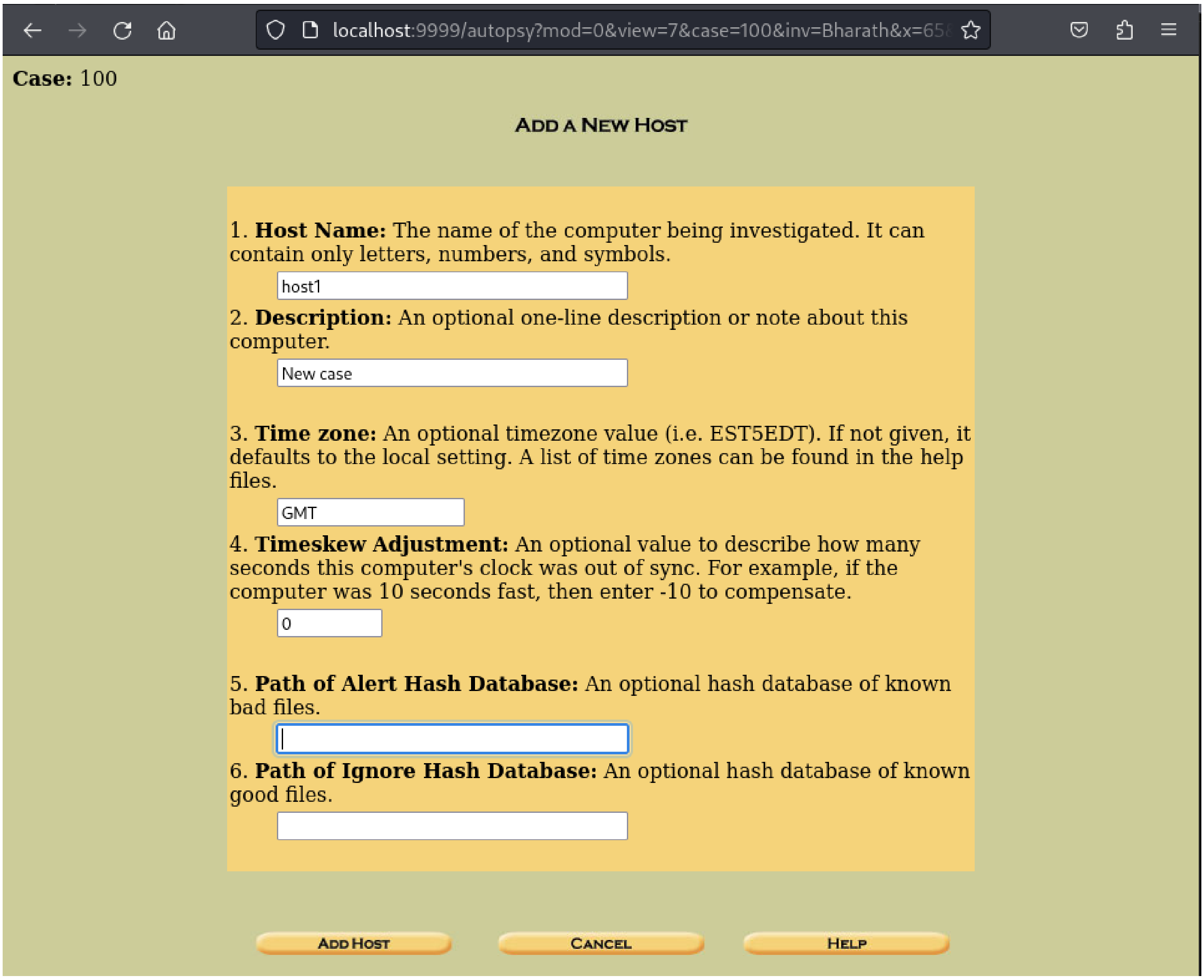
3. Fill the required details in the **CREATE NEW CASE** page then click **NEW CASE.**



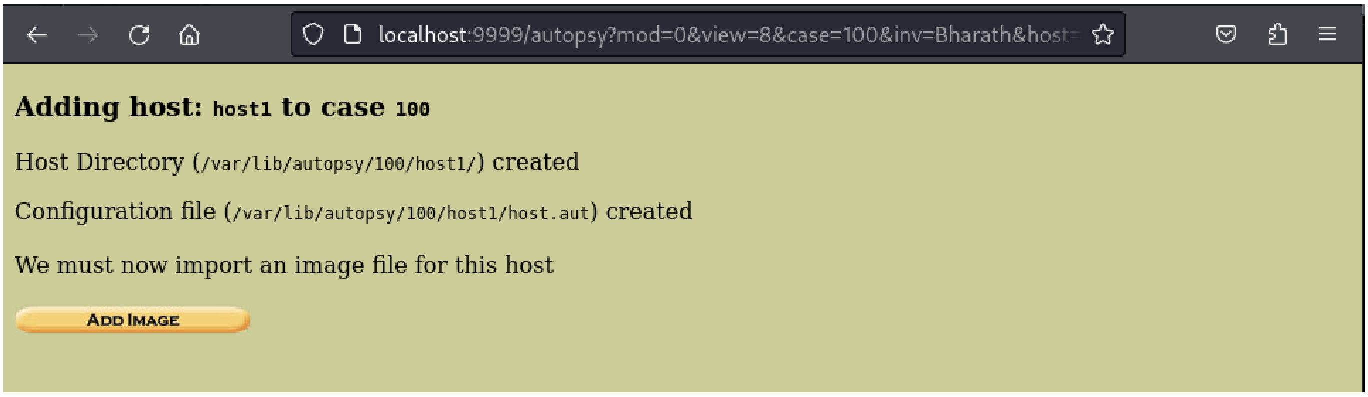
4. Now click **ADD HOST** button



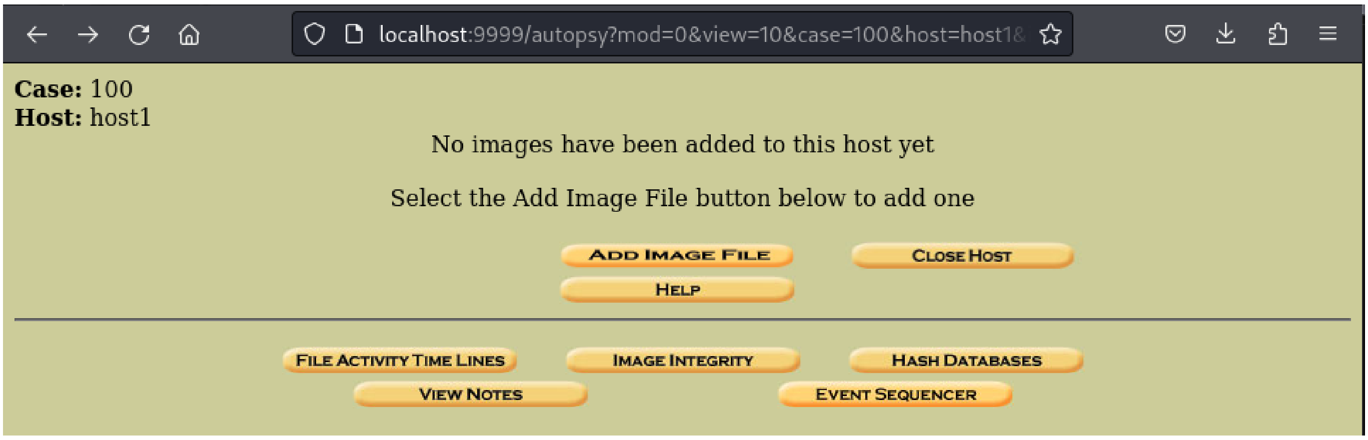
5. Fill the details in the **ADD A NEW HOST** page and click **ADD HOST** button.



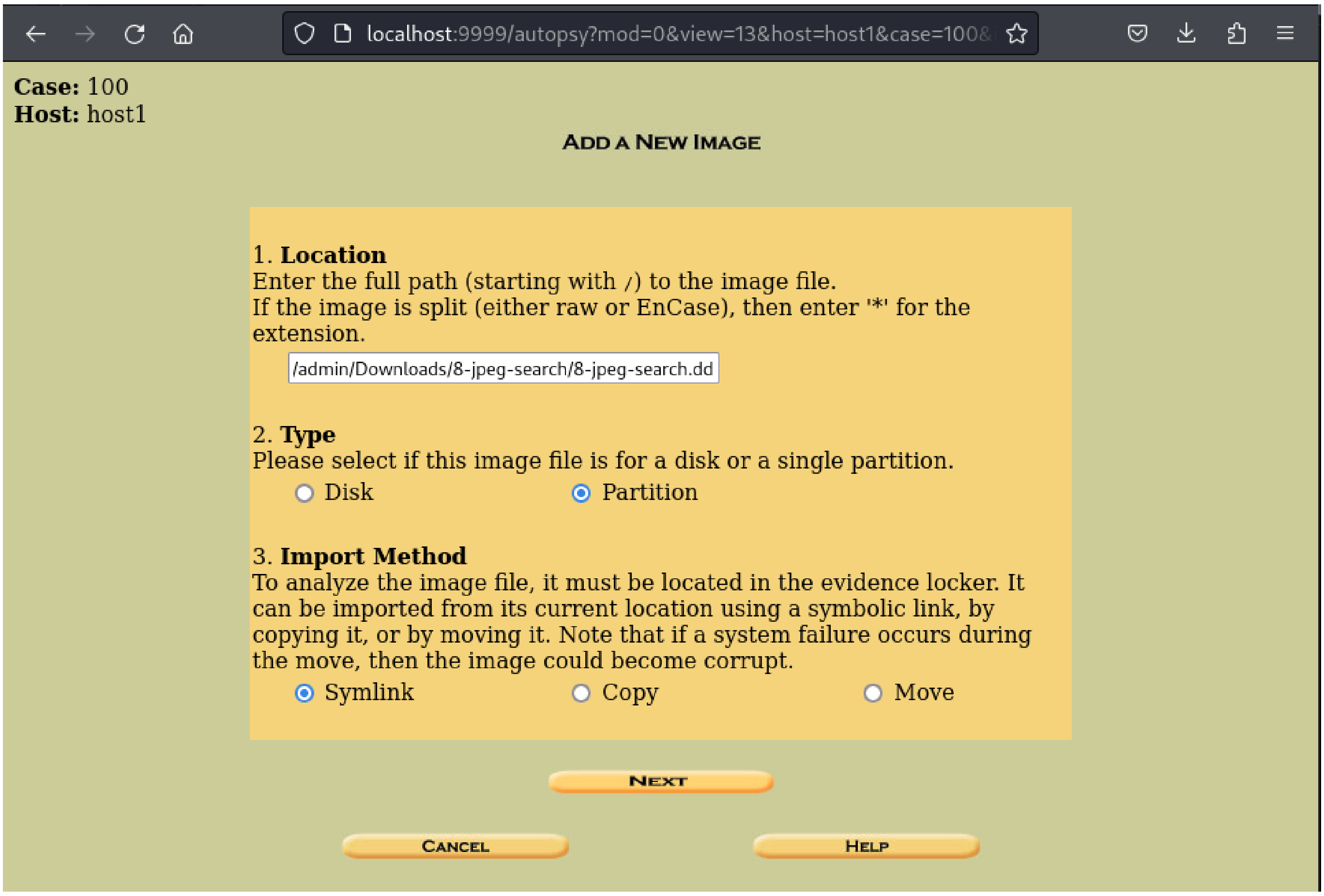
6. Click **ADD IMAGE** button



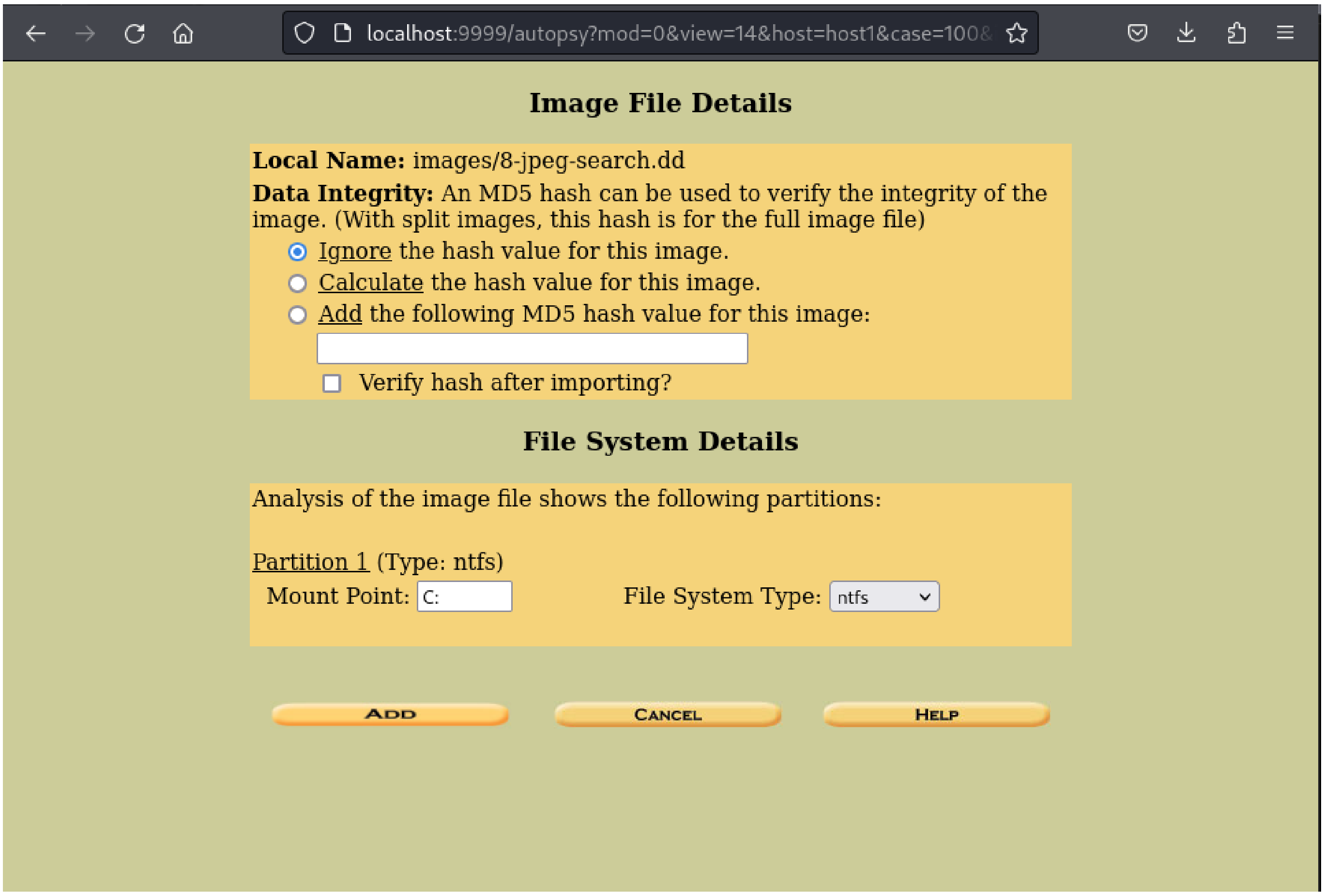
7. Click **ADD IMAGE FILE** button to add an image for investigation



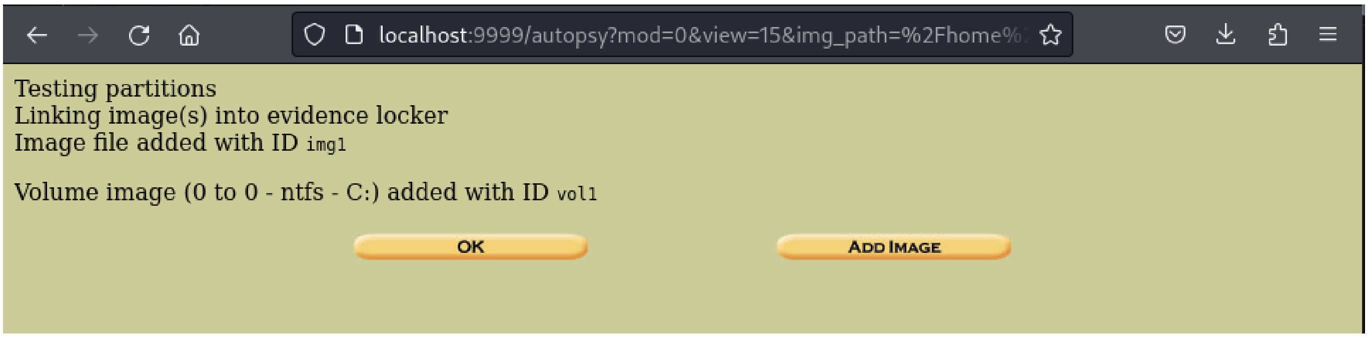
8. Enter the image file location and select image type then click next.



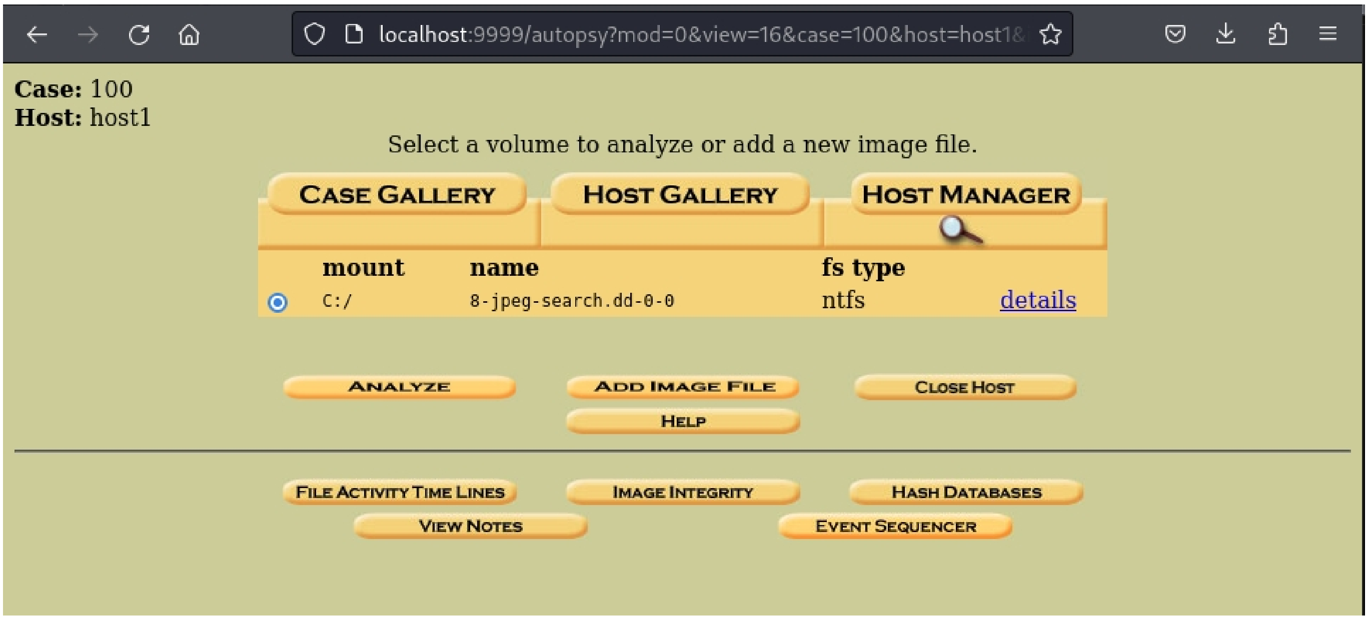
9. Click **ADD** button in the Image file details page.



10. Testing Partition page appears , click OK.

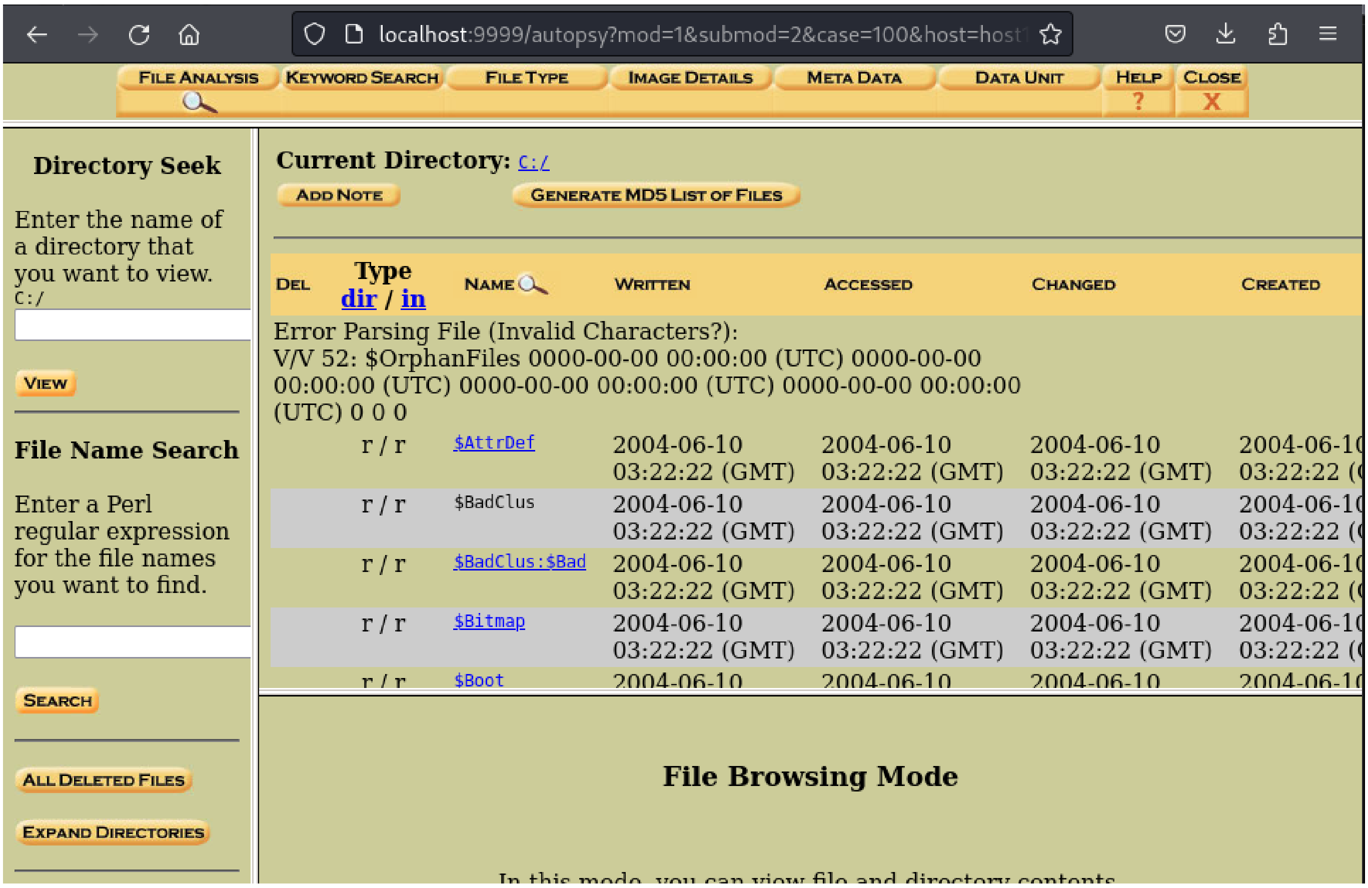


11. Once the image added to the autopsy database, you can analyze the image. To anlyze the image click **ANALYZE**



12. To do the file analysis, click **FILE ANALYSIS** button that allows you to analyze the image from the file and directory prespective.

13. File Analysis is used to examine the directories and files for evidence.



14. To generate MD5 hashes of the contained files , click **GENERATE MD5 LIST OF FILES** button.